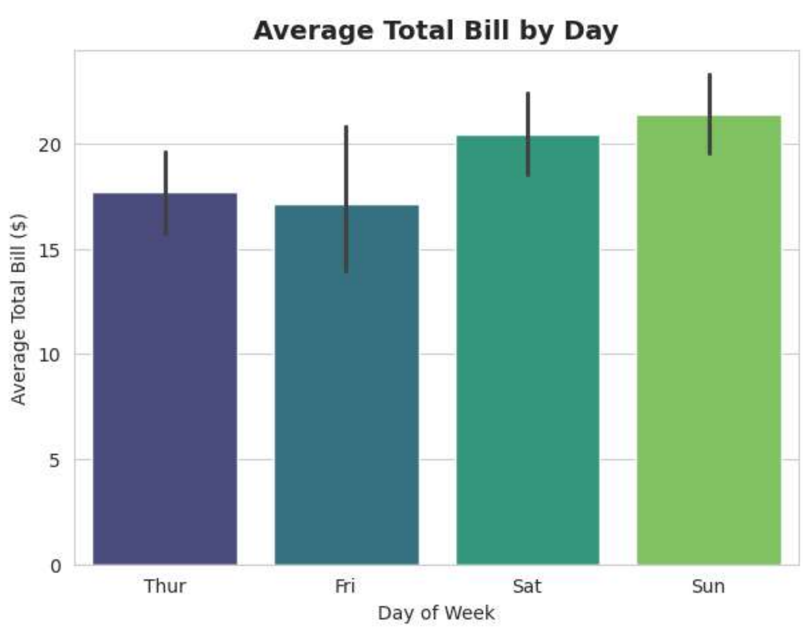
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.set_style("whitegrid")
```

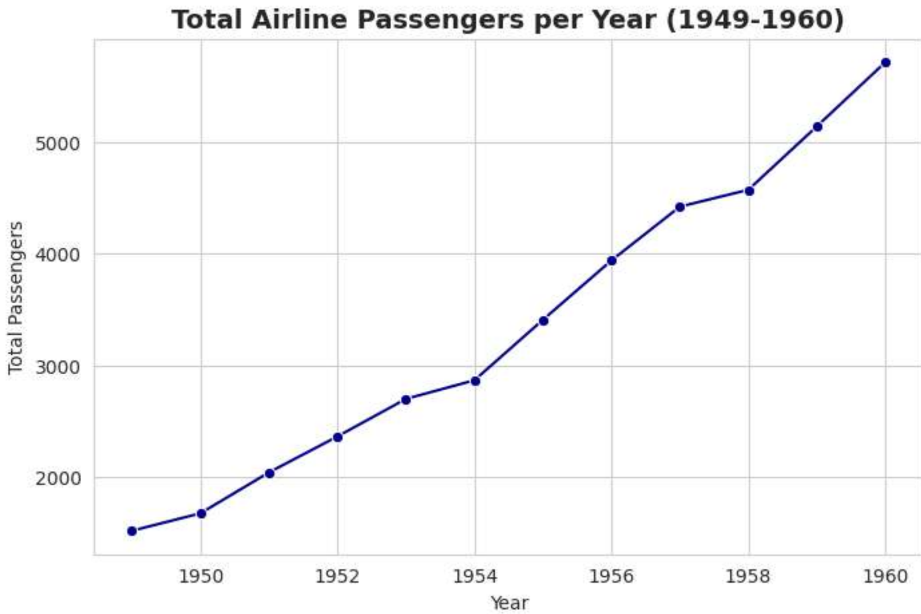
```
tips = sns.load_dataset("tips")
print(tips.head())
print(tips.info())
```

```
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   total_bill  244 non-null    float64
 1   tip         244 non-null    float64
 2   sex         244 non-null    category
 3   smoker      244 non-null    category
 4   day         244 non-null    category
 5   time        244 non-null    category
 6   size        244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
None
```

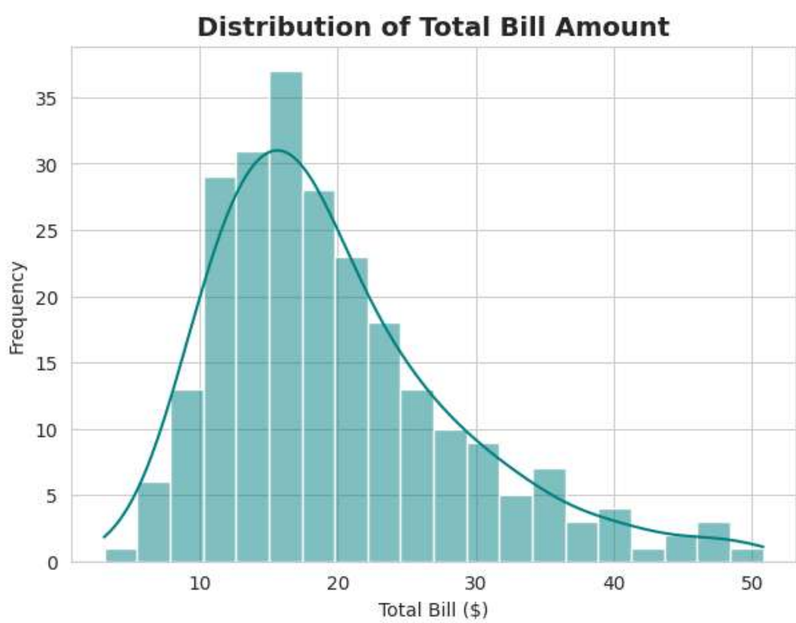
```
plt.figure(figsize=(7, 5))
sns.barplot(data=tips, x="day", y="total_bill", hue="day", palette="viridis", legend=False)
plt.title("Average Total Bill by Day", fontsize=14, fontweight="bold")
plt.xlabel("Day of Week")
plt.ylabel("Average Total Bill ($)")
plt.show()
```



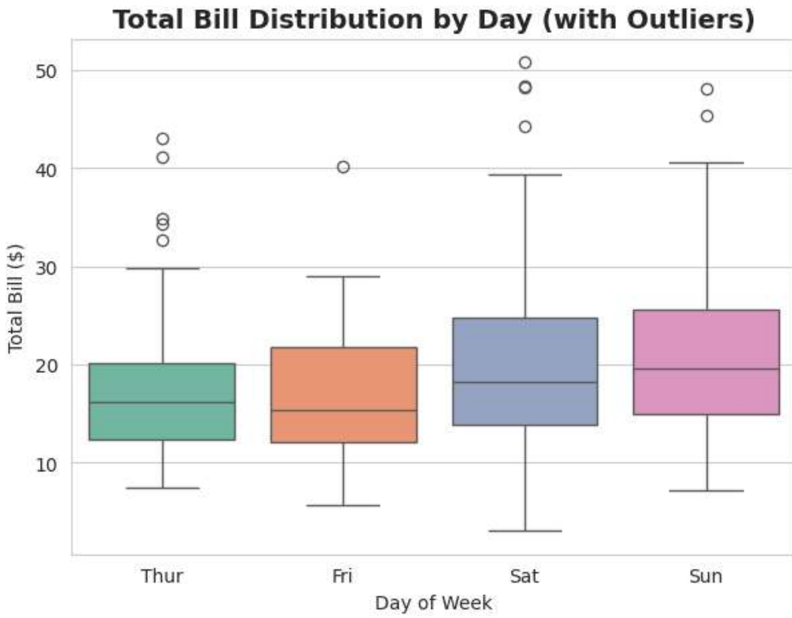

```
flights = sns.load_dataset("flights")
yearly = flights.groupby("year")["passengers"].sum().reset_index()
plt.figure(figsize=(8, 5))
sns.lineplot(data=yearly, x="year", y="passengers", marker="o", color="darkblue")
plt.title("Total Airline Passengers per Year (1949-1960)", fontsize=14, fontweight="bold")
plt.xlabel("Year")
plt.ylabel("Total Passengers")
plt.show()
```



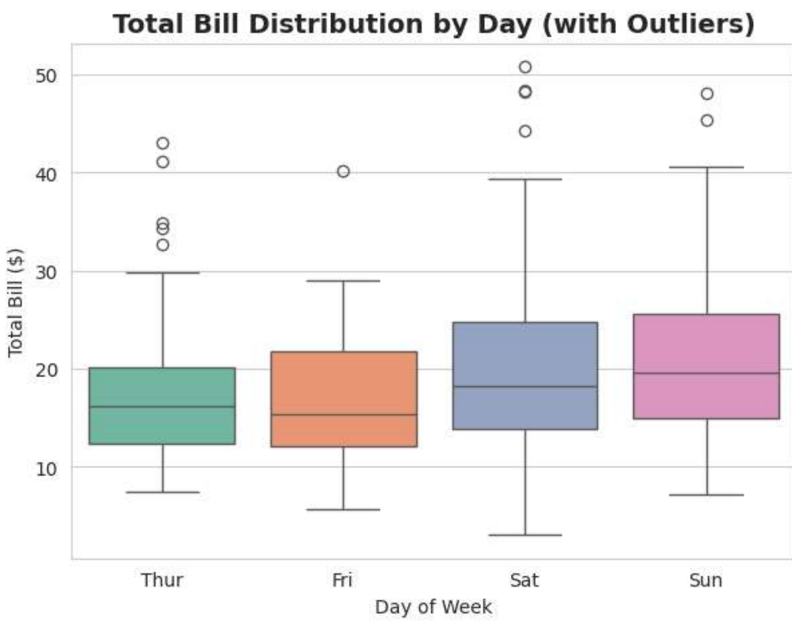

```
plt.figure(figsize=(7, 5))
sns.histplot(tips["total_bill"], bins=20, kde=True, color="teal")
plt.title("Distribution of Total Bill Amount", fontsize=14, fontweight="bold")
plt.xlabel("Total Bill ($)")
plt.ylabel("Frequency")
plt.show()
```



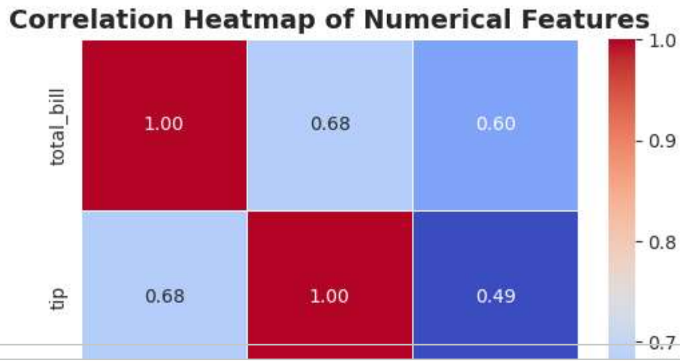

```
plt.figure(figsize=(7, 5))
sns.boxplot(data=tips, x="day", y="total_bill", hue="day", palette="Set2", legend=False)
plt.title("Total Bill Distribution by Day (with Outliers)", fontsize=14, fontweight="bold")
plt.xlabel("Day of Week")
plt.ylabel("Total Bill ($)")
plt.show()
```




```
plt.figure(figsize=(7, 5))
sns.boxplot(data=tips, x="day", y="total_bill", hue="day", palette="Set2", legend=False)
plt.title("Total Bill Distribution by Day (with Outliers)", fontsize=14, fontweight="bold")
plt.xlabel("Day of Week")
plt.ylabel("Total Bill ($)")
plt.show()
```




```
numeric_cols = tips.select_dtypes(include="number")
corr = numeric_cols.corr()
plt.figure(figsize=(6, 5))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap of Numerical Features", fontsize=14, fontweight="bold")
plt.show()
```




```python
plt.figure(figsize=(8, 5))
ax = sns.barplot(data=tips, x="day", y="tip", hue="day", palette="magma", legend=False)
plt.title("Average Tip by Day of Week", fontsize=16, fontweight="bold", color="darkred")
plt.xlabel("Day", fontsize=12)
plt.ylabel("Average Tip ($)", fontsize=12) # Add value annotations on top of each bar for bar in ax.patches:    height = bar.get_
```

```
Text(0, 0.5, 'Average Tip ($)')
```